

# Algebra: The Distance Formula

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## DIRECTIONS

Use the Distance Formula  $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$  to find the distance between each pair of points. Simplify all radicals.

|                                                           |                                                                                                         |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1. Find d:<br><b>(0, 0) and (3, 4)</b><br>Answer: _____   | 2. Find d:<br><b>(1, 2) and (4, 6)</b><br>Answer: _____                                                 |
| 3. Find d:<br><b>(-3, 0) and (0, 4)</b><br>Answer: _____  | 4. Find d:<br><b>(2, -1) and (2, 5)</b><br>Answer: _____                                                |
| 5. Find d:<br><b>(-1, -1) and (2, 3)</b><br>Answer: _____ | 6. Find d:<br><b>(0, -3) and (4, 0)</b><br>Answer: _____                                                |
| 7. Find d:<br><b>(-2, 3) and (4, -5)</b><br>Answer: _____ | 8. Find d:<br><b>(3, -2) and (-1, 1)</b><br>Answer: _____                                               |
| 9. Find d:<br><b>(-5, 2) and (3, -4)</b><br>Answer: _____ | 10. Which is farther? A: (0,0)→(6,8) or B: (-3,1)→(3,-7)<br><b>Show work for both.</b><br>Answer: _____ |

Based on the Numberbender lesson "ALGEBRA: Using the Distance Formula" • [youtu.be/5JilGerPyW4](https://youtu.be/5JilGerPyW4)

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# Answer Key & Solutions

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## TEACHER NOTES

Items 1–6: results are perfect squares — no radical simplification needed. Items 7–8: results simplify cleanly. Items 9–10: compare or apply distances. Remind students to square both differences before adding under the radical.

1. Find d: (0, 0) and (3, 4)

**Answer: 5**

$$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5.$$

2. Find d: (1, 2) and (4, 6)

**Answer: 5**

$$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5.$$

3. Find d: (–3, 0) and (0, 4)

**Answer: 5**

$$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5.$$

4. Find d: (2, –1) and (2, 5)

**Answer: 6**

$$\sqrt{(0^2+6^2)} = \sqrt{36} = 6.$$

5. Find d: (–1, –1) and (2, 3)

**Answer: 5**

$$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5.$$

6. Find d: (0, –3) and (4, 0)

**Answer: 5**

$$\sqrt{(4^2+3^2)} = \sqrt{(16+9)} = \sqrt{25} = 5.$$

7. Find d: (–2, 3) and (4, –5)

**Answer: 10**

$$\sqrt{(6^2+8^2)} = \sqrt{(36+64)} = \sqrt{100} = 10.$$

8. Find d: (3, –2) and (–1, 1)

**Answer: 5**

$$\sqrt{(4^2+3^2)} = \sqrt{(16+9)} = \sqrt{25} = 5.$$

9. Find d: (–5, 2) and (3, –4)

**Answer: 10**

$$\sqrt{(8^2+6^2)} = \sqrt{(64+36)} = \sqrt{100} = 10.$$

10. Which is farther? A: (0,0)→(6,8) or B: (–3,1)→(3,–7) Show work for both.

**Answer: Equal — both d = 10**

A:  $\sqrt{(36+64)}=10$ ; B:  $\sqrt{(36+64)}=10$ . Same distance.

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